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	FINAL EXAMINATION COVER PAGE	**Issue No.** 1 / **Page No.** Page 1 of 1

Mathematics Department of the College of Natural and Social Science

Final Examination for the Course Mathematics for Natural Sciences (Math1007)

43

Examination Date: Aug 08, 2022

Maximum score: 50%

Time Allowed: 2 hours and 45 minutes

Tick the box, where it applies you.

Freshman	Applied Science	☐	Engineering	☑
Regular:	Undergraduate	☑	Postgraduate	☐
Continuing Education	Undergraduate	☐	Postgraduate	☐

Student's name: Abel Tessema Mulugeta

ID No: ETS 0027/14

Section: A1

If you are adding this course, please tick the box ☐

General Directions

- Check that there are **seven** pages including the cover page in this closed book examination booklet.
- Check that there are **six** multiple choice, **seven** fill the blank space and **five** work-out questions.
- Attempt all the questions according to the instruction of each part.
- Use the backside of the pages for any rough work.
- Strictly forbidden materials to use:
  - **Any Calculators, mobile phone, extra papers, red pen and pencil.**
- Cheating is strictly forbidden and automatically nullifies your total scores.

DO NOT START UNTIL YOU ARE TOLD TO DO SO!

WISH YOU ALL THE BEST!!

PLEASE MAKE SURE THAT THIS IS THE CORRECT ISSUE BEFORE USE

5+12+26

Instruction I. Choose the choice with the right answer and write the letter of your choice on the shaded space provided ONLY (1 points each)

A 1. The possible complex number $z = a + bi$ such that $z\bar{z} = 25$ and $a + b = 7$ is
A. $2+5i$ B. $5+2i$ C. $1+6i$ D. $4+3i$

C 2. If the eccentricity values of four ellipses with the same major axis length are 0.1, 0.5, 0.8 and 0.3 , then the ellipse closest to be the circle is the ellipse with eccentricity of
A. 0.5 B. 0.3 C. 0.1 D. 0.8

D 3. The exact numerical value of $\cosh(-\ln 3) + \sinh(\ln 2)$ is ___.
A. $\frac{11}{6}$ B. $\frac{13}{9}$ C. $\frac{5}{4}$ D. $\frac{29}{12}$

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D 4. For propositions p, q and r, which pair of compound propositions are not equivalent?
A. $p \Rightarrow q; \neg p \vee q$ B. $r \Rightarrow \neg q; q \Rightarrow \neg r$ C. $\neg(r \Rightarrow \neg p); p \wedge r$ D. $\neg(\neg q \Rightarrow \neg p); q \Rightarrow p$

B 5. Which one has truth value false in the set of real numbers?
A. $(\forall x)(\forall y): (x+y)^2 \geq 2xy$ C. $(\forall x)(\exists y): x^2 - y^2 = x + y$
B. $(\exists x)(\forall y): y - x = x - y$ D. $(\exists x)(\exists y): x^2 + y^2 = x + y$

C 6. Which of the following is not true about the conic section $4x^2 - 24x - 25y^2 + 250y = 489$?
A. Center is (3,5) B. conjugate axis length is 10 C. one focus is (3,9) D. one vertex is (3,7)

Instruction II: Fill the following blank spaces by the most simplified and finalized answers on the given shaded spaces ONLY. (2 points each)

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1. The domain of $f(x) = \sqrt{5x - x^2}$ is [0, 5].
2. For two sets A and B if $n(A \setminus B) = n(B \setminus A) = 4$ and $n(A \cap B) = 3$, $n(A \cup B)$ = 11.
3. The point of tangency of a circle with center lying on the line $y = 4x$ is (1, 0); its radius is $\frac{4\sqrt{17}}{17}$
4. The LCM and GCF of two numbers respectively are $2^3 \times 3^2 \times 5 \times 7$ and 6. If one of the numbers is $3^2 \times 2 \times 7$, then the other number is 120.
5. The principle argument of $\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)$ is $\frac{\pi}{2}$
6. When $f(x) = x^4 + 3x^2 - 8$ is divided by $x - 2$, the remainder is 20
7. If $2 + 4i$ is one root of the equation $x^2 + bx + c = 0$ then the value of c is 20

Instruction III: Work out the following questions showing all the necessary steps clearly on the given space provided (6 points each)

1. Consider the argument,

   If Ethiopia is the origin of mankind, then Ethiopia is in East Africa. It is not true to say that either Ethiopia is in East Africa or Afar is in Ethiopia. Therefore, if Ethiopia is not the origin of mankind, then Afar is not in Ethiopia.

   i. Write the above argument using logical operators (2 points)

   ii. Check the validity of the argument using or without using truth table (4 points)

i) Let p: Ethiopia is the origin of mankind.
q: Ethiopia is in East Africa.
r: Afar is in Ethiopia.

Thus, the argument can be written as:

$p \Rightarrow q, \neg(q \vee r) \vdash \neg p \Rightarrow \neg r$

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ii) Check:

① $\neg(q \vee r)$ is true -- Hypothesis

② $q \vee r$ is false --- Negation of ①

③ 'q' is false & 'r' is also false --- From ② & the rule of disjunction.

④ $p \Rightarrow q$ is true --- Hypothesis

⑤ 'p' is false --- From ③, ④, and the rule of inference

⑥ $\neg p$ is true --- Negation of ⑤

⑦ $\neg r$ is true --- Negation of ③

⑧ $\neg p \Rightarrow \neg r$ is true --- From ⑥, ⑦, and the rule of inference.

By assuming that our premises are true, we always get a conclusion which is also true.

Ergo, it is valid.

OR

Let's make a truth table:

p	q	r	$\neg p$	$\neg r$	$p \Rightarrow q$	$q \vee r$	$\neg(q \vee r)$	$\neg p \Rightarrow \neg r$
T	T	T	F	F	T	T	F	T
T	T	F	F	T	T	T	F	T
T	F	T	F	F	F	T	F	T
T	F	F	F	T	F	F	T	T
F	T	T	T	F	T	T	F	F
F	T	F	T	T	T	T	F	T
F	F	T	T	F	T	T	F	F
F	F	F	T	T	T	F	T	T

Only on the last row are both premises true. And on that row, the conclusion is also true.

Since it's impossible for the premises to be true yet the conclusion false, the argument is valid.

2. Use principle of mathematical induction to prove that

$6^n - 1$ is multiple of 5, $\forall n \in N$ = set of natural numbers

Proof:

Step 1:- For $n=1$, $6^1 - 1 = 5$ which is a multiple of 5.

Step 2:- Assume that it is true for $n=k$, i.e.,

$6^k - 1$ is a multiple of 5, or $6^k - 1 = 5m$ for $m \in \mathbb{Z}$

Step 3:- We should show that it is true for $n=k+1$

Assumption on Step 2 $\Rightarrow$ $6^k - 1 = 5m$ for $m \in \mathbb{Z}$

$\Rightarrow 6^k = 5m + 1$,

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Now, $6^{k+1} - 1 = 6 \cdot 6^k - 1$

$= 6(5m+1) - 1$

$= 30m + 6 - 1$

$= 30m + 5$

$= 5(6m+1)$

$5(6m+1)$ is a multiple of 5, since 5 is its factor.

Hence, $6^{k+1} - 1$ is a multiple of 5.

We have proved that it is true for $n=k+1$ whenever it is true for $n=k$.

Therefore, it is true for all natural numbers 'n'.

3. Given a rational function $f(x)=\dfrac{2x^2+3}{x-1}$, find its

a) domain (1 point)

b) x and y intercepts if any (1 point)

c) All possible existing asymptotes if any (2 points)

d) Sketch the graph (2 points)

6

a) $D_f = \mathbb{R} \setminus \{1\}$

b) $x_{int} \Rightarrow 0 = \dfrac{2x^2+3}{x-1} \Rightarrow 2x^2+3=0 \Rightarrow 2x^2=-3 \Rightarrow$ No x-intercept

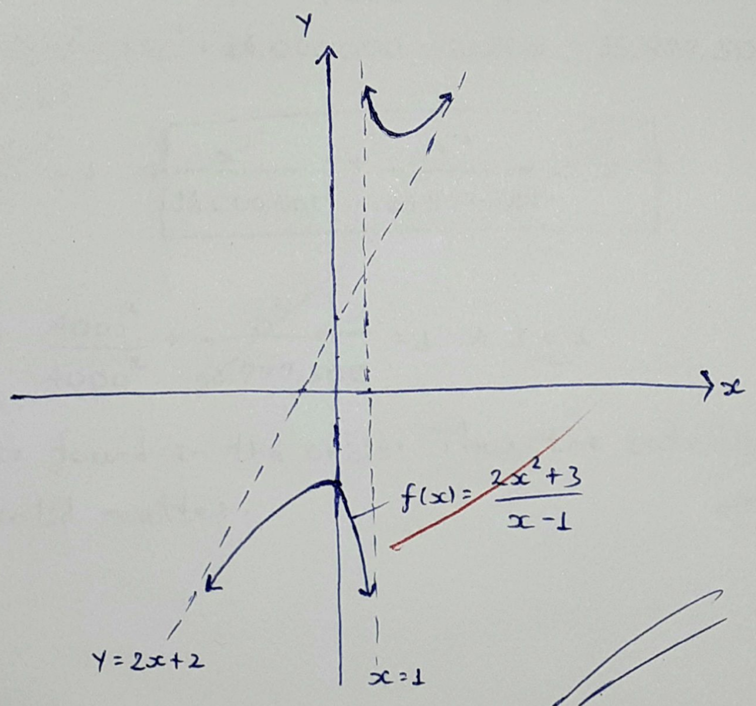
$y_{int} \Rightarrow y = \dfrac{2(0)^2+3}{0-1} = -3 \Rightarrow (0,-3)$

c) V.A $\Rightarrow x-1=0 \Rightarrow x=1$

H.A $\Rightarrow$ None

O.A $\Rightarrow$ $x-1 \,\overline{)\,2x^2+3}$ with quotient $2x+2$: $-(2x^2-2x)$, $2x+3$, $-(2x-2)$, 5 $\Rightarrow y = Q(x) \Rightarrow y = 2x+2$

d)



$x-1 \quad 2x^2+0x+3$

$\begin{array}{c|ccc} 1 & 2 & 0 & 3 \\ & & 2 & 2 \\ \hline & 2 & 2 & 5 \end{array}$

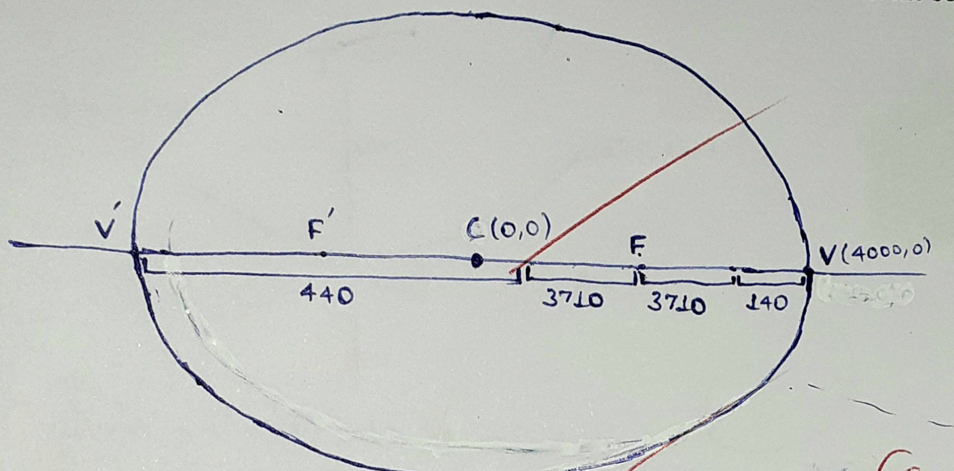
$2x+2,\ R=5$

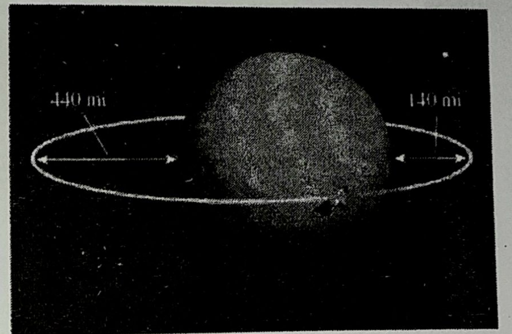
$\dfrac{\frac{1}{2}+3}{-\frac{1}{2}} = \dfrac{\frac{7}{2}}{-\frac{1}{2}} = -7$

4. A satellite is an elliptical orbit around the earth with the center of the earth at one focus, as shown in the figure. The height of the satellite above the earth varies between 140 and 440. (All units in miles). Assume that the earth is a sphere with radius 3710. (Place the coordinate axes so that the origin is at the center of the satellite orbit and the foci are located on the x-axis).

i. Find the equation of the elliptical satellite's orbit? (4 points)

ii. If a solid matter is stationed permanently at coordinate of (4000, 0) with the given coordinate axes. Do you think the satellite collide with the solid matter? (Justify your answer) (2 points)





It's not quite representatively accurate, but it'll do for our calculations.

6

i) $2a = 440 + 3,710 \times 2 + 140 = 8,000 \Rightarrow a = 4,000$

$\overline{CV} = \overline{CF} + \overline{FV} \Rightarrow a = c + \overline{FV} \Rightarrow 4,000 = c + 3710 + 140 \Rightarrow c = 150$

$b^2 = a^2 - c^2 = 4000^2 - 150^2 = 16,000,000 - 22500 = 15,977,500$

$C(h,k) = C(0,0)$

$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1 \Rightarrow \boxed{\frac{x^2}{16,000,000} + \frac{y^2}{15,977,500} = 1}$

ii) $P(\overset{x}{4000}, \overset{y}{0}) \Rightarrow \frac{4000^2}{4000^2} + \frac{0^2}{15,977,500} = 1 \Rightarrow 1 = 1$

The point is found in the orbit. Thus, the satellite collides with the solid matter.

$\frac{x^2}{4000} + \frac{y^2}{b^2} = 1$ $\quad x-h \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

3710 × 2 = 7420, 580, 8000

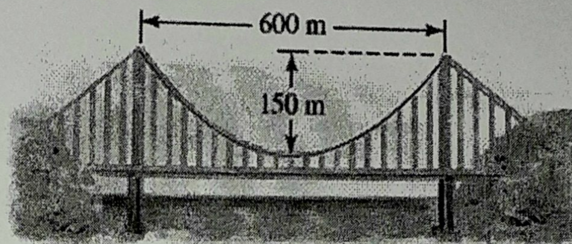
16,000,000 − 22,500 = 15977,500

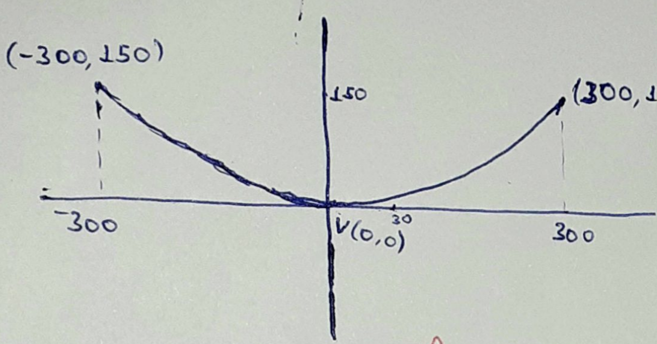
3710 + 140 = 3850 ⇒ 4000 − 3850 = 150

5. In a suspension bridge the shape of the suspension cable is parabolic. The bridge shown in the figure has towers that are 600m apart, and the lowest point of the suspension cable is 150m below the top of the towers. (Placing the origin of the coordinate system at the vertex)

i. Find the equation of the parabolic cable? (4 points)

ii. What is the height of the cable 30m away from a tower? (2 points)





2

i) $(x-h)^2 = 4P(y-k) \Rightarrow x^2 = 4PY$

Let's take the point $(x, y) = (300, 150)$

$300^2 = 4P\,150 \Rightarrow P = \frac{300\times300}{4\times150} = 150$

Thus, the equation is: $\boxed{x^2 = 150Y}$

ii) $x = 30$, $Y = ?$

$x^2 = 150Y \Rightarrow Y = \frac{x^2}{150} = \frac{30^2}{150} = \frac{30\times30}{150} = \frac{300\times3}{150} = 6$ m